UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES

BSc General Degree First Year First Semester Course Unit Examination – October/November 2022

DEPARTMENT OF COMPUTER SCIENCE

ICT 105 1.5 Computer Architecture

Time: One (01 1/2) hours | No. of questions: 04 | No. of pages: 02 | Total marks: 100

Answer ALL questions

1. i. Suppose that there are 2^{30} bytes in a memory. What are the lowest and highest memory addresses, if memory is byte addressable? How many bits are needed for the address? (15 marks)

   ii. Perform the following arithmetic operations in 6-bit registers using signed-2's complement form for negative numbers. Indicate if it is overflowing or not. (10 marks)

   (a) $(14)_{10} + (18)_{10}$

   (b) $(-14)_{10} + (18)_{10}$

   (Total: 25 marks)

2. i. What are the main characteristics of RISC processors? (5 marks)

   ii. A program runs in 10 seconds on computer A, which has a 400MHz clock. A new machine B, that would run the same program in 6 seconds, has to be designed. The designer can use new (or perhaps more expensive) technology to substantially increase the clock rate, but has informed that this increase would affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for the same program. What should be the clock rate of B? (5 marks)

   iii. The computer *A* and *B* has to be compared. The computer *A* has the machine instructions for the floating-point operations, while *B* has not (FP operations are implemented in the software using several non-FP instructions). Both computers have a clock frequency of 400 MHz. In both the same program is executed, which has the following mixture of commands:

Type the command	Dynamic share of instructions in program (p_i)	Instruction duration (Number of clock periods CPI_i)	
		A	*B*
FP addition	16%	6	20
FP multiplication	10%	8	32
FP division	8%	10	66
Non-FP instructions	66%	3	3

   (a) Calculate the MIPS for the computers *A* and *B*.

   (b) Calculate the CPU program execution time on the computers *A* and *B*, if there are 12000 instructions in the program.

   (c) At what mixture of instructions in the program will both computers *A* and *B* be equally fast? (15 marks)

   (Total: 25 marks)

3. i. Translate the following high-level code in C language into MIPS or RISC-V assembly language. Assume variables a–c are held in registers $s0–$s2 and f–j are in $s3–$s7.

```
a = b - c;
f = (g + h) - (i + j);
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(15 mar

ii. Translate the following machine language code into MIPS assembly language. (10 mar

0x2237FFF1

(Total: 25 mar

4. i. Briefly explain the functionalities of buses used to specify processors. (10 mar

ii. Briefly describe the Fetch-Decode-Execute cycle. (15 mar

(Total: 25 mar

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